Design of the Civil Works for Singapore's North East Line

R. Krishnan, J. P. Copsey, R. G. Algeo, and J. N. Shirlaw

Abstract *—Singapore's North East line is a 20-km-long subway line, including 16 underground stations. The civil works were let in 1997 on a design / build basis, with 12 major civil contracts and one trackwork contract. Most of the stations are designed as civil defence shelters. Provision has been made for future developments over many of the stations, and for future road, rail and drainage projects. The paper summarises some of the design requirements for both temporary and permanent works design, and comments on how these have been met in practice.* © 2000 Published by Elsevier Science Ltd. All rights reserved.

1.0 Introduction

The current, operational Singapore Mass Rapid Transit system has been constructed in phases between 1984 and 1997. Of the 48 existing stations, 15 are underground and 33 at grade or on viaduct (Fig. 1). In January 1996, the Singapore Government authorised the construction of a new line, the North East Line (NEL). Tender documents were issued later in the same year, and the first construction contract was awarded in April 1997. This line, consisting of 16 underground stations linked by bored and cut-and-cover tunnels, runs from the World Trade Centre to Punggol (Fig. 1). The overall length of the line is 20 km. The work has been divided into 12 major construction packages, ranging in value from 85/m to 350/m Singapore dollars (approximately $US50.5/m to $US206/m). These 12 civil contracts do not include the trackwork, which has been let as a separate contract.

The initial alignment of the North East line was established in a study carried out in 1986. However, significant modifications were made before tenders were called in 1996. The 12 main civil contracts were awarded between April 1997 and March 1998 on a design-build basis. The Land Transport Authority of Singapore issued, as part of the tender documents, drawings showing the planned horizontal and vertical alignment, and the location and general footprint of the stations. Limited modification to both the alignment and the station footprint by the contractors was allowed, within the railway boundaries established by the Authority.

At the time of writing (August 1999), the civil design of the North East line is complete, and construction is about 33% complete. Excavation for the stations is largely complete, with the formation of the station shells in progress. Tunnelling is also in progress, with 14 shield machines operating. The line is expected to open, as originally scheduled, in 2002.

Present addresses: R.Krishnan, Land Transport Authority, No 1, Hampshire Road, Singapore 219428; and J.P. Copsey, R.G. Algeo, and J.N. Shirlaw, Land Transport Authority, No 1, Hampshire Road, Singapore 219428

This paper will present some of the civil design issues, and, in particular, the Authority's design requirements and how they have been met.

2.0 Ground Conditions

To investigate the ground conditions, the Authority drilled 535 boreholes and carried out 127 Cone Penetration Tests along the route of the North-East Line.

The geology along the North-East Line is presented in Figure 2, which shows the route superimposed on the Geological map of Singapore (PWD 1976). The route runs through four major geological units: the Bukit Timah granite, the Jurong Formation, the Old Alluvium, and the Kallang Formation.

The *Bukit Timah Granite* is the oldest of these units, and is a term used to cover a suite of granitic rocks, predominantly microgranite and granodiorite. The *Jurong Formation* consists of sedimentary rocks of Jurassic age. These rocks have been heavily folded through being pushed against the Bukit Timah Granite, and consist of interbedded mudstone, siltstone and sandstone.

The *Old Alluvium* is the youngest of the rock formations; Gupta et al. (1987) identify it as being of "possible Pleistocene age". It was laid down when sea level was up to 146 m below current sea level. The Old Alluvium consists largely of mechanically weathered granite eroded from the mountains of what is now Indonesia, and deposited into a huge drainage system between Singapore/Malaysia and the Island of Borneo. The alluvial deposit has become weakly lithified with age, and much of it can now be classified as very weak to weak sandstone or siltstone.

The rocks of the Bukit Timah Granite, Jurong Formation and the Old Alluvium have been subjected to tropical weathering, typically to depths of 10 m to 60 m.

The *Kallang Formation* consists of soils of Holocene and late Pleistocene age. These include fluvial sands and clays, but most commonly are soft marine and estuarine clays. These clays are near normally consolidated and highly compressible.

Rainfall in Singapore averages 2.4 m per year, with the monthly rainfall typically varying between 170 mm and 260 mm. The Island of Singapore has little topographic relief.

www.elsevier.com/locate/tust

Tunnelling and Underground Space Technology, Vol. 14, No. 4, pp. 433–448, 1999
0886-7798/00/ $ - see front matter © 2000 Published by Elsevier Science Ltd. All rights reserved.
PII: S0866-7798(00)00006-7



Pergamon

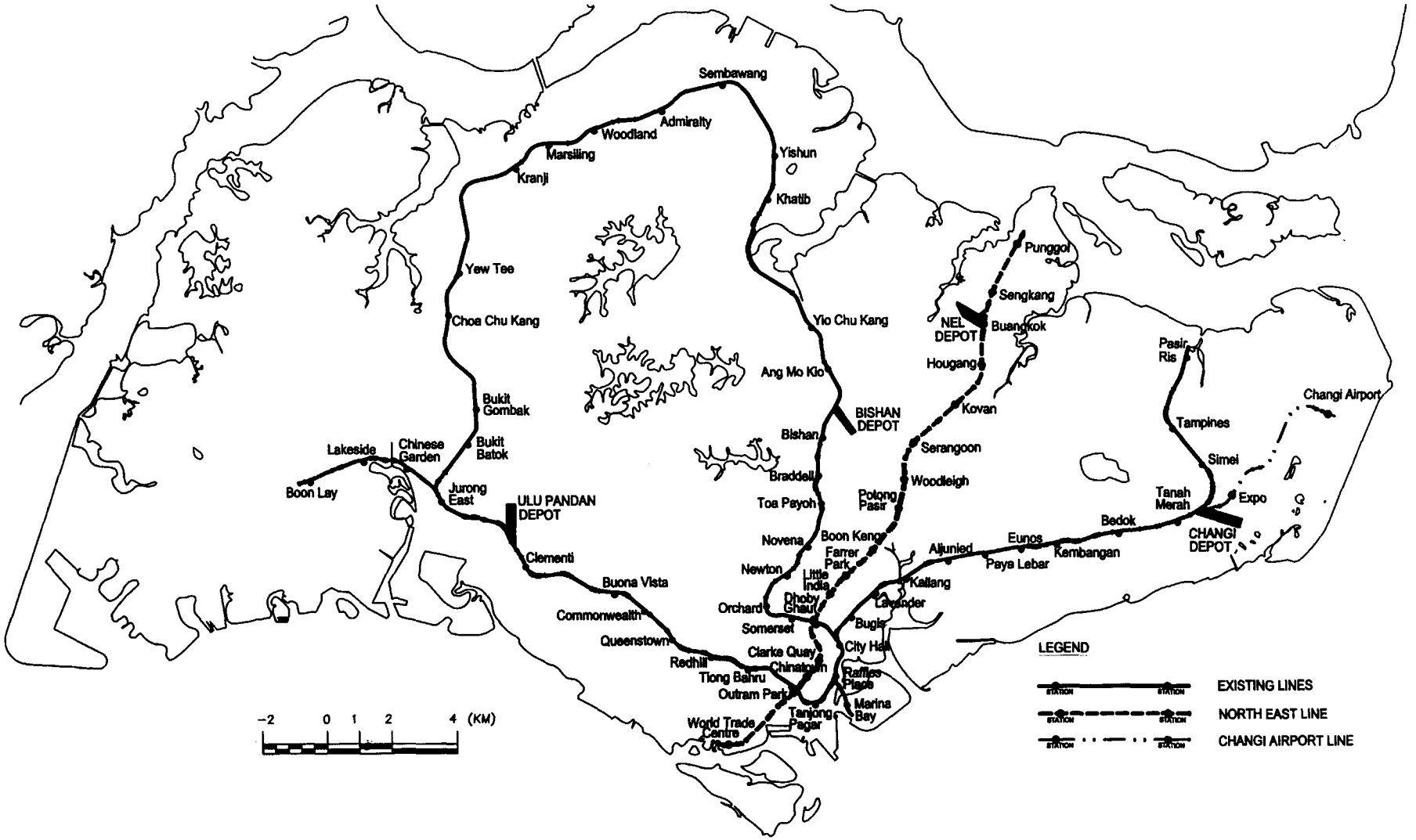


Figure 1. Singapore subway line.

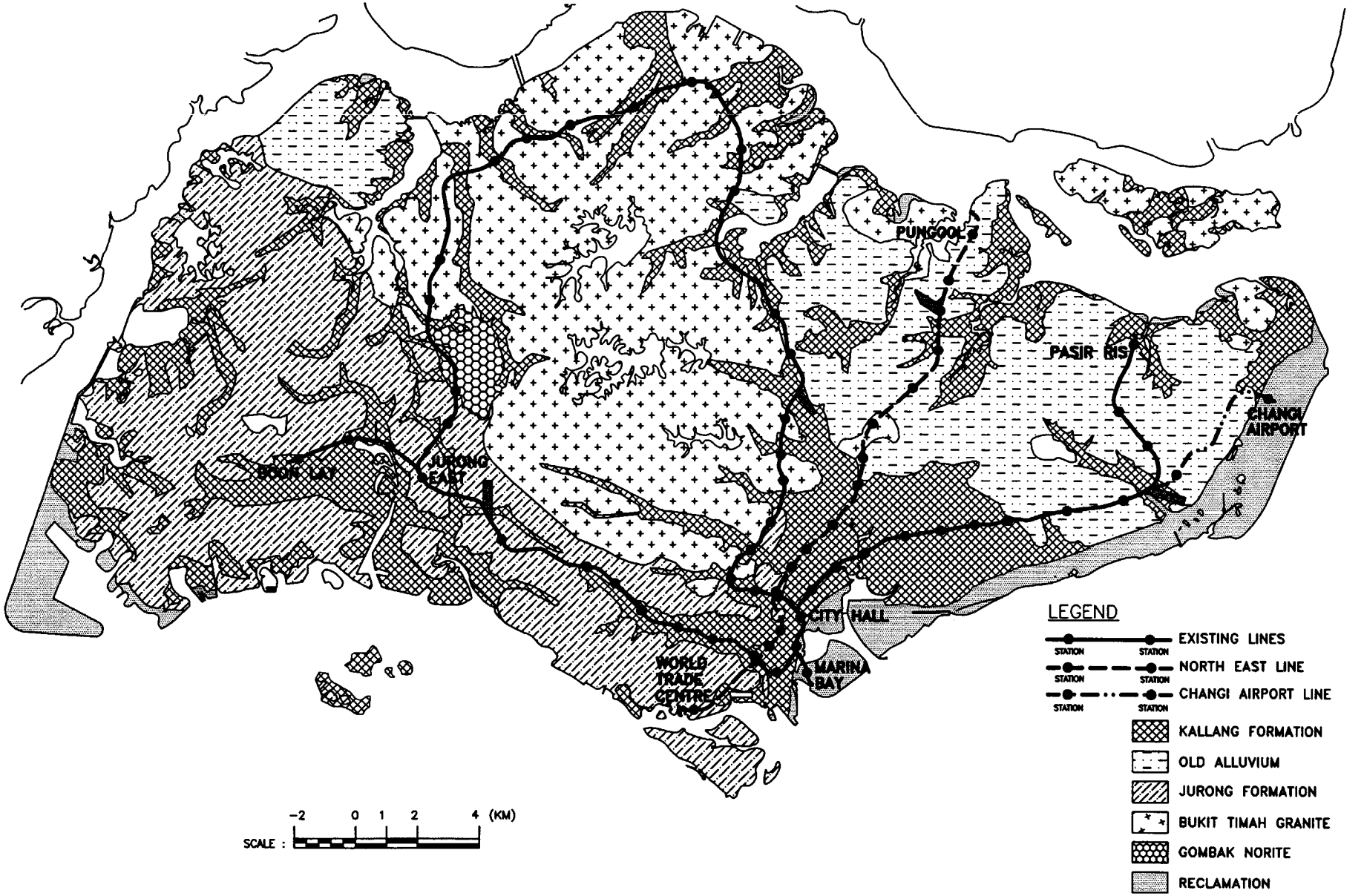


Figure 2. The Singapore subway lines superimposed on the geological map.

Much of the older urban area is built on land which is just 5 m above mean sea level. The highest point on the island is Bukit Timah hill, at 161 m, while "Mount" Sophia is just 25 m above mean sea level. The combination of high rainfall and limited relief results in a ground water level which is commonly 1 m to 2 m below ground level, and tends to follow the local topography. For underground construction, a reduced level of +100 m is used for mean sea level; this allows all of the work to be above +0 m and obviates the risk of mistakes between positive and negative levels.

While the general geology of the North East Line can be obtained from the geological map of Singapore, the site investigation was necessary to more precisely define subsurface conditions for the construction of stations and tunnels. Information obtained from all of the Authority's boreholes has been summarised in Figure 3.

From Figures 2 and 3, it can be seen that the North-East Line divides conveniently into 3 distinct zones. The southern zone, from World Trade Centre to Little India Stations, is the area where the main rock type in the Jurong Formation. Deep deposits of Kallang Formation soils and fill are found locally in this zone, particularly in the Singapore River and Chinatown area.

The central zone, between Little India and Potong Pasir Stations, is in the area known as the "Kallang Basin". This area is characterised by deep deposits of soils of the Kallang Formation. Because the North East Line passes across the western edge of the basin, the depth of the Kallang formation soils encountered was highly variable. Through this area the line passes through a number of buried valleys incised into the underlying Old Alluvium, and infilled with marine clay and fluvial sand of the Kallang Formation. The maximum depth of the Kallang Formation deposits has been at Farrer Park Station, with a depth to Old Alluvium of 37 m from ground surface.

The northern zone of the route, from Woodleigh to Punggol Stations, is predominantly in the Old Alluvium. However, there is a buried ridge of variably weathered Bukit Timah Granite at Serangoon Station and along the adjacent sections of bored tunnel. Generally there is only a thin layer of fill over the Old Alluvium in the northern part of the route. However, small valleys infilled with soils of the Kallang Formation to a depth of up to 15 m below ground level are apparent locally.

A few of the tenderers for the work elected to drill a limited number of additional boreholes before bidding the work, to supplement those provided by the Authority. After the award of the main construction contracts, all of the successful tenderers carried out a supplementary investigation programme. Typically this programme was carried out to refine key geotechnical parameters for the design of the work, or to cover areas such as entrances or crosspassages where the extent or location were not fully defined at the tender stage.

2.1 Design Criteria

The design of the North East line is specified to follow Singapore Standards, as established by the Productivity and Standards Board. As the number of Singapore Standards is relatively limited, the appropriate British Standard is specified where there is no Singapore Standard. In the absence of an appropriate Singapore or British Standard, other standards published by a "reputable institution" may be proposed.

As well as complying with the relevant standard, the designs must comply with the Design Criteria developed by the Authority. These Design Criteria cover a wide variety of topics that are specific to the design of the underground railway and work in Singapore. For example, one of the two volumes of the Design Criteria relates only to the design of Stations as Civil Defence shelters.

Topics within the Design Criteria range from the loadings imposed by the subway trains in operation to the signage required during temporary road diversions. It is not possible to discuss all of the requirements here. However, some requirements pertinent to this paper are outlined below.

- **Ground water levels**

The ground water level for all permanent works design is specified to be full hydrostatic pressure from ground level.

- **Use of diaphragm or secant pile walls as part of the permanent works**

It was not acceptable to have exposed diaphragm walls or secant piles as the final surface of the stations or tunnels. Internal walls had to be provided, either cast directly against the external walls or (in stations) with cavity wall construction. In the assessment of the civil defence requirements, the full thickness of any diaphragm wall used can be considered in satisfying the requirement for the minimum thickness of the external wall. However, if secant pile construction is used, not more than half the thickness of secant pile wall can be considered.

- **Anchors**

All ground anchors have to be destressed and the strand removed after use.

- **Flotation**

Because of the high design groundwater levels, design against flotation is often critical. However, use of the friction between the structure and the soil is permitted. Partial factors are applied to the weight of the structure (1.1), weight of undisturbed soil (1.15), weight of backfill (1.3) and frictional resistance (2.0). After taking into account these partial factors, an overall factor of safety against flotation of 1.15 is required.

- **Design for adjacent developments**

Within the limited land area of Singapore, it has to be expected that there will be extensive redevelopment along the route, often involving deep excavations close to the stations and tunnels. Work close to the railway has to be in accordance with the "Code of Practice for Railway Protection" (CPRP). The CPRP requires that movement of the existing transit structure due to adjacent development is less than 15 mm in any plane, and rotation of the rails is less than 1:2000 in any plane. As these movements are considered permissible under the CPRP, the Design Criteria require that the NEL structures are designed for imposed movements of this magnitude following construction.

- **Settlement and adjacent buildings**

The contractors are required to predict ground movements resulting from the construction of the works and assess the effects of those movements on buildings and utilities. It is specified that the assessment shall include the calculation of horizontal tensile strain in the buildings, as outlined in Boscardin and Cording (1989) and Mair, Taylor and Burland (1996). Buildings which are predicted to suffer "moderate" damage or greater (based on the categories established by Burland, Broms and de Mello (1977) were to be protected, to reduce the damage level to "slight" or less. In the case of a few buildings of historic importance, damage was to be limited to "very slight".

- **Tunnel lining design (structural)**

The segmental linings were specified to be 5.8 m internal diameter minimum. The lining is specified to be designed for full overburden except in fresh rock, where reduced loads

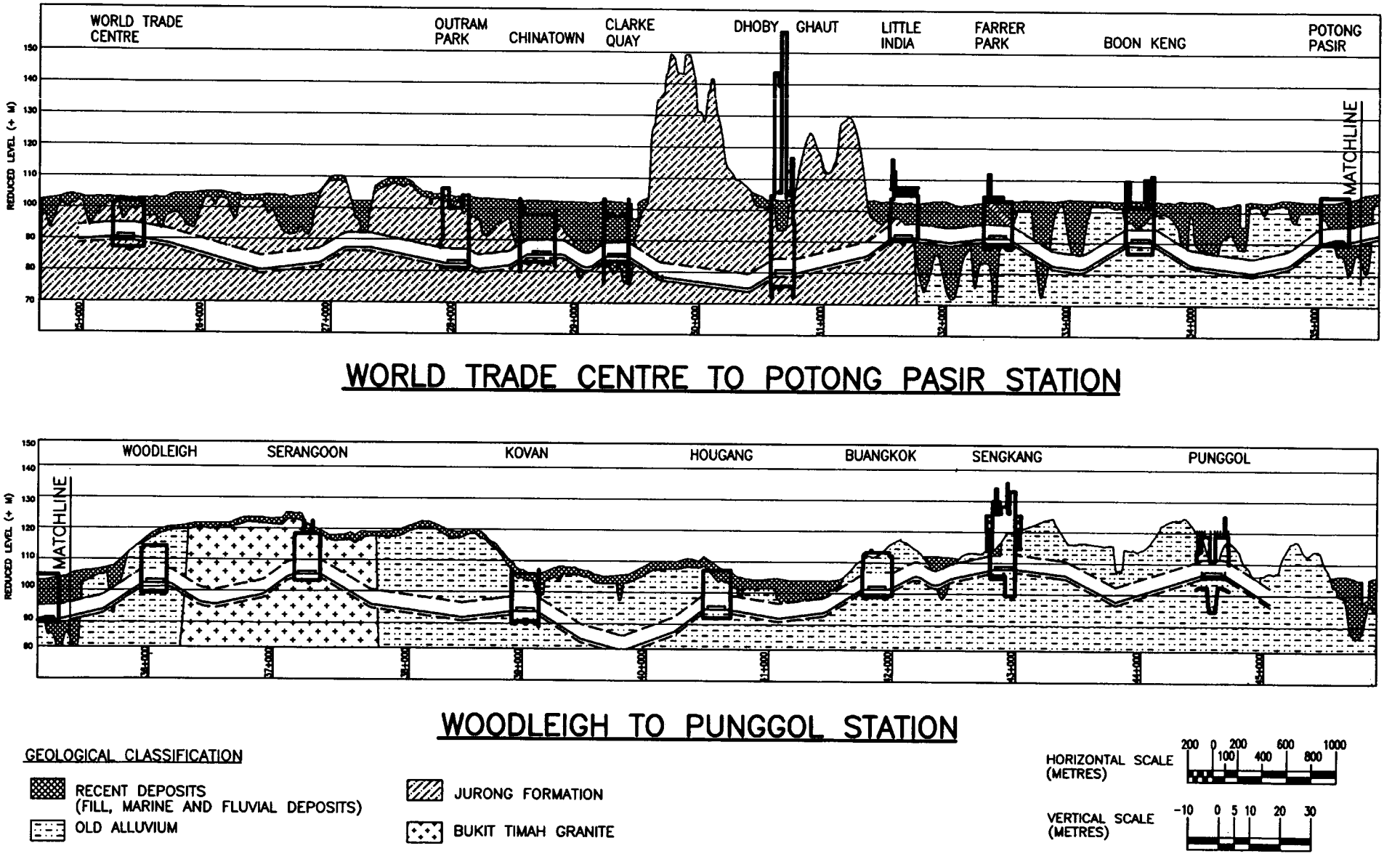


Figure 3. A long section showing the stratigraphy along the North East Line.

can be used. The design was required to satisfy the moments and thrusts derived from a plane strain continuum method based on the work of Muir Wood (1975) and Curtis (1976), with distortional loading coefficients based on the experience of the previous MRT construction.

- **Tunnel lining design (durability)**

Many of the rivers and canals which pass over the NEL are tidal, and relatively high levels of salinity were measured in the groundwater, particularly in the southern part of the NEL. Experience from the existing MRT lines had shown that a combination of low permeability concrete with a coal tar epoxy coating on the outer face provided an effective barrier to the passage of groundwater through the body of the segments. However, corrosion damage to the existing linings was apparent, particularly due to ingress of water at segmental lining joints. So, while the requirements for low-permeability concrete and a coal tar epoxy coating (to all but the inner face and caulking groove) were maintained, additional requirements for gaskets at the joints were introduced.For the NEL, it was required that two gaskets were provided—one EPDM and one hydroswelling. The tender documents also specified the use of epoxy coated rebar, but this requirement was withdrawn following discussions with the tenderers, and replaced by criteria relating to the permeability of the concrete.

2.2 Design Process and Review

Under Singapore law, all designs have to be undertaken by a Professional Engineer (PE) registered in Singapore. Major structural elements are also subject to an independent check, which has to be carried out by an Accredited Checker (AC). The construction has to be supervised by a Qualified Person (QP), who is also responsible for confirming the suitability of the temporary works design.

In addition to the reviews under the PE, AC and QP requirements, the contractors' designs were subject to review both by the Authority and a number of bodies with statutory powers.

The contractors' designs are reviewed by the Authority for conformance to the Authority's requirements and specifications. For the permanent works three major submissions were required. These submissions, the preliminary, pre-final and final submission were required at typically 6, 12 and 18 months after tender award. The design of temporary works also had to be submitted for review, but the timing was only specified to be such that the Authority could review it and appropriate changes made before installation.

In addition to review by the Authority, the contractors had to allow for submission and review by a number of bodies with statutory powers. These included:

- Development Control Division of the Urban Redevelopment Authority, for planning permission for all railway structures except tunnels
- The Building Control Unit of the Land Transport and Construction Authority, who are authorised by the Building Authority of Singapore to issue building plan approval for railway structures. In order to obtain building plan approval, the applicant has to be able to certify that the requirements of various "Technical Departments" have been satisfied. Depending on the nature of the work, these "Technical Departments" include: Sewerage, Drainage, Environmental Health, Pollution Control Department, Public Works Department, National Parks Board, Singapore Power Ltd, Water Department, and the Civil Defence Shelter Bureau.
- The Development and Building Control Unit of the Land Transport Authority, who give permission for work within the "protection zone" of the existing Mass Rapid Transit system.

Although two of these bodies were groups within the Land Transport Authority, their roles and responsibilities are quite distinct from that of the Authority's staff directly supervising the design and construction of the North East line. In practice they act as independent reviewers within a clearly defined scope for their review.

3.0 Permanent Works Design

The permanent works schemes for the 16 stations of the North-East Line and for the cut-and-cover tunnels are summarised in Table 1 and Table 2, respectively.

All of these stations are cast-insitu reinforced concrete box-type structures. They are constructed either in open cut excavations, or in strutted excavations retained by soldier piles and lagging, sheetpiles or diaphragm or secant pile walls.

Thirteen of the stations are also designed as civil defence shelters. For these structures the member sizes are frequently dictated by minimum criteria given in the Civil Defence requirements. The details are elaborated in section 3.1, below.

Principal loadings are for (future) oversite developments (where these need to be catered for), surcharge loading, soil and hydrostatic pressure, CD debris loading (where applicable), and plant loading (on internal slabs).

Analysis is generally by 2-dimensional frame analysis of typical sections, supplemented by grillage analyses of slabs, where appropriate. In some instances, and particularly where large point loads occur (due. for example. to future developments), 3-dimensional analyses of the station structures have been carried out to examine both the transverse and longitudinal behaviour of the structure.

Design is to BS8110, with crackwidth due to flexure and to early thermal cracking limited to 0.2 mm.

Durability is required to meet the "extreme" exposure condition of BS8110:1985, and the design life was specified as 100 years.

All structures are waterproofed, generally by a bonded membrane system. In base slabs, waterproof concrete incorporating a pore-blocking additive has been adopted in some locations in lieu of a bonded membrane.

A key planning objective has been to open up the stations by minimising the number of columns in the platform areas. In many stations, platform edge columns are provided along the line of the platform screen doors. In some cases, these support only the concourse floor above, while in others they provide support to every station's slab and also to the future developments.

3.1 Civil Defence Issues

3.1.1 Shape and size

The general shape and configuration of the NEL shelter stations and their locations are controlled by peacetime transit requirements. As such, shelter stations typically have long concourse and platform areas with several connecting access ways.

Most of the NEL stations have just two levels: a concourse level and a platform level. However, the two interchange stations (Dhoby Ghaut and Outram Park) have up to five underground levels because of the depth needed to allow the NEL tunnels to pass under the existing tunnels. CD shelter requirements are provided either by locating the shelter outside the railway area (at Dhoby Ghaut) or by designing only the lower concourse and platform levels of the station (at Outram Park) as a CD shelter.

3.1.2 Protection scenario

The NEL shelter stations are designed to operate as transit stations during peacetime and as CD shelters during emergency. As shelters, they are designed to withstand

Table 1. A summary of the temporary and permanent works for the North East Line Stations.

Station Name	Civil Defence (CD) or Non-CD	Provision for Loads from future oversite development	Track Level(TL) Base Soffit Level (SL)	Depth to underside of base slab (m)	Construct -ion Sequence	Temporary Works for Excavation	Permanent Works Scheme	Foundations
Punggol	Non-CD	LRT under concurrent construction	+104 TL +102.67 SL	18.3	Bottom up	Open cut.	Cast insitu reinforced concrete box	Bored piles
Sengkang	Non-CD	4 storey HDB development and LRT under concurrent construction	+105.50 TL +103.6 SL	8.9	Bottom up	Open cut	Reinforced concrete beams and slabs supported on RC columns.	Bored piles
Buangkok	CD	None	+99.00 TL +95.925 SL	16.6	Bottom up	Open cut	Reinforced concrete slabs supported on RC walls.	Base slab designed as raft foundation
Hougang	CD	Provision for future HDB development comprising : 13 sty residential or 9 sty office,1 sty commercial + 2 sty carpark,1 sty bus interchange	+92.350 TL +89.15 SL	19.3	Bottom up	Northern End : Open Cut Southern End : Strutted excavation using diaphragm wall	Diaphragm wall + Cast insitu RC composite wall	Combination of raft foundation and piled foundation: piles designed to resist tension
Kovan	CD	None	+91.500 TL +88.25 SL	21.7	Bottom up	Strutted excavation using diaphragm wall	Diaphragm wall + Cast insitu RC composite wall	Base slab designed as raft foundation
Serangoon	CD	1) Road viaduct (under same Contract). 2) Future 6-storey HDB development above Entrance 2.	+103.64 TL +99.9 SL	24.1	Bottom up	Strutted excavation using soldier piles & timber lagging	Cast insitu reinforced concrete box structure	Base slab designed as a raft foundation
Wood-leigh.	CD	None	+100.64 TL +96.9 SL	23.1	Bottom up	Strutted excavation using soldier piles & timber lagging	Cast insitu reinforced concrete box structure	Base slab designed as a raft foundation
Potong Pasir	CD	None	+88.00 TL +85.90 SL	17.8	Top down	Top down - roof slab supported by diaphragm walls and plunge columns	Cast insitu reinforced concrete box structure with diaphragm walls and inner insitu walls designed as composite member	Diaphragm walls and bored piles. Some parts of the base slab also acts as raft
Boon Keng	CD	None	+88.00 TL +85.90 SL	17.2	Top down	Top down - roof slab supported by diaphragm walls and plunge columns	Reinforced concrete box structure with diaphragm walls and inner insitu walls designed as composite	Combined foundation of diaphragm walls with base slab acting as raft.

continued on next page

Table 1, continued.

Station Name	Civil Defence (CD) or Non-CD	Provision for Loads from future oversite development	Track Level(TL) Base Soffit Level (SL)	Depth to underside of base slab (m)	Construct -ion Sequence	Temporary Works for Excavation	Permanent Works Scheme	Foundations
Farrer Park	CD	Provision for future URA loads	+89.00 TL +86.55 SL	16.4	Bottom up	Diaphragm wall with strutted excavation	Reinforced concrete box structure with diaphragm walls and inner insitu walls	Bored piles and diaphragm walls
Little India	CD	Provision for future URA loads	+89.00 TL +85.75 SL	17.2	Bottom up	Diaphragm wall with strutted excavation	Reinforced concrete box structure with diaphragm walls and inner insitu walls	Bored piles and diaphragm walls
Dhoby Ghaut	Non-CD (CD facility in car-park)	Provision for future URA loads: URA Development at north and south ends of Dhoby Ghaut Station	+77.5 TL +74.5 SL	30 to 39	Bottom up	Soldier piles, sheet piles, ground anchors, rock bolts, secant piles, struts, shotcrete	RC box	Base slab acts as a raft foundation
Clarke Quay	CD	Provision for future URA loads	+82.42 TL +79.8 SL	22.6	Bottom up	Diaphragm wall with strutted excavation	Diaphragm wall + RC walls & slabs	Combination of diaphragm wall, barrettes and base slab acting as raft
China-town	CD	None	+83.00 TL +80.70 SL	21.5	Bottom up	Secant pile wall with strutted excavation	Reinforced concrete box structure with secant pile and inner insitu walls.	Base slab designed as a raft foundation. Entrances supported on bored piles
Outram Park	CD	Entrance 1 : 10-15 storey URA future development. Entrance 4 : 6-15 storey URA future development	+80.0 TL +77.0 SL	26.5	Bottom up	Strutted excavation using soldier piles. Secant piles and ground anchors need where required.	RC box	Base slab as a raft foundation, Entrance 1 and end of linkway on piles to carry future development loads.
World Trade Centre	CD	Provision for future PSA development loads	+88.08 TL +84.95 SL	18.5	Bottom up	Strutted excavation using sheet pile/soldier pile walls and timber lagging	RC box	Base slab designed as a raft foundation

Table 2. A summary of the temporary and permanent works for the cut-and-cover tunnels of the North East Line.

Adjacent Station(s)	Provision for Loads from future oversite development	Backfill thickness above tunnel (m)	Base Soffit Level	Construction Sequence	Temporary Works for Excavation	Permanent Works Scheme	Foundations
Punggol	Loads from LRT under concurrent construction	Minimum 3m (below Tampines Expressway) Maximum 10 m	96.00 (av.)	Bottom up	Generally open cut, locally soldier piles and lagging where space precludes open cut	Cast insitu reinforced concrete box	Base slab designed as raft
Sengkang &Buangkok	Loads from LRT under concurrent construction	Varies from 2 m to 8 m.	Varies from 102.60 to 90.30	Bottom up.	Open cut & strutted sheet pile excavations	Cast in-situ reinforced concrete structures.	Base slab designed as raft
Hougang	Provision for HDB development at southern end of the tunnel	Typically 8m	88.379	Bottom up	Southern End : Open Cut The rest : strutted excavation using sheetpiles	Cast insitu reinforced concrete box	Combination of raft foundation and piled foundation
Potong Pasir	One of the foundations of a future foot bridge will be supported by diaphragm wall.	Varies from 9m to 11 m.	Varies from 88.77 to 86.82	Top down	Strutted excavation using diaphragm walls	Cast insitu reinforced concrete box, with diaphragm walls and inner insitu walls designed as composite member.	Diaphragm walls and bored piles in certain areas, with ground stiffness taken into account for raft action.
Little India & Farrer Park	Provision for future URA loads	7.8m	85.8	Bottom up	Strutted excavation using diaphragm walls	Cast insitu reinforced concrete box with diaphragm walls	Diaphragm walls with bored piles in some areas
Outram Park	None	17m	77.8	Bottom up	Strutted excavation using soldier piles, waling and struts	Cast insitu reinforced concrete box	Base slab as raft
World Trade Centre	1. **Cross Over Tunnel:** Provision for future PSA development loads	≈ 9m	Approx 85.0	Bottom up	Strutted excavation using sheetpiles/soldier piles walls/timber lagging	Cast insitu reinforced concrete box	Raft foundation
	2. **Sidings Tunnel:** Provision for future PSA/Keppel development loads	≈ 8-9 m	Approx 85.0 to 83.3	Bottom up	As above	As above	As above

weapon effects including blast, shock, fragmentation and unhealthy air caused by smoke, dust and other products of explosive combustion. This involves structural hardening and necessitates that the shelter station be airtight and be equipped with air filtration facilities. This in turn determines the system of air supply and exhaust for the shelters.

3.1.3 Zones and bounds of protection

Just as a transit station is classified into paid and unpaid areas, public and non-public areas, the shelter station is classified into several distinct zones of protection. The zones are demarcated by their respective bounds of protection. These bounds are defined by walls and slabs, including the CD doors and CD valves in them. The thickness of the walls and slabs and the type of CD doors and CD valves must complement the bounds of protection which they serve.

3.1.4 Structural requirements

The structural design of the shelter stations has to ensure that the shelters perform satisfactorily under various loading conditions. Unlike conventional peacetime structures, the shelters have to be subjected to both peacetime design requirements, and to CD load conditions which result from weapon effects. Three main areas that impact the structural design of the shelter station are presented here.

3.1.5 Minimum thickness and reinforcement

Minimum thickness and reinforcement requirements for various structural members are to be complied with.

While the civil defence minimum member thickness requirements for the external hull generally control over peacetime requirements, this is not always the case for deep shelters and high water tables. For such shelters, foundation slabs deeper than 15 m below adjacent finished ground level may be designed to peacetime requirements. For high water tables, the shelter hull may have to be thickened to counter flotation effects. On the other hand, for internal members the minimum member thickness for semi-protected, blast and internal (shock) members generally exceeds peacetime requirements.

The amount of reinforcement to be provided for a member depends on the load case. In particular, reinforcement design and detailing in the case of external hull members must be based on special "cubic lattice" arrangements as prescribed in the CD Design Criteria. The reinforcement provided has to be checked against minimum requirements prescribed.

3.1.6 Additional load cases

In addition to peacetime load cases, the shelter station has to be designed to CD load cases. These are the debris load case, extraordinary static load case and dynamic load case.

a) Debris Load Case

Debris loads result from the destruction of nearby buildings above and around the shelter station. Where the adjacent buildings are designed against progressive collapse, then the debris load is taken as 20 kN/m^2 and is applied vertically onto the station roof slab and horizontally onto the station external wall. Otherwise, the debris load is taken as 50 kN/m^2.

The debris load application is in addition to peacetime loads and, as reduced load factors are permitted, its application may or may not present a more onerous load case depending on the depth of the shelter station. Generally, the deeper the station, the less critical the debris load.

b) Extra-Ordinary Static Load Case

The extra-ordinary static load (EOSL) case applies to the design of the external shelter hull. It assumes localised failure of individual load bearing members such as columns and piles. Hence, the shelter station roof slab and foundation slab have to be designed such that any one of its columns or pile groups can be allowed to fail. This ensures robustness of the station hull.

Again, the EOSL case may not be more critical than the peacetime load case. This depends on the number of and distance between the supporting columns and pile groups. But it is an important load condition to be considered.

c) Dynamic Load Case

Blast loads and shock loads on the shelter station are dynamic in nature. These loads are usually large but of short duration.

Structural members that constitute the bounds of blast protection, or the bounds of air and blast protection, are subjected to blast loads, e.g. the walls of blast locks and the walls separating the ventilation shafts. Internal structural members such as the concourse and platform slabs are subjected to shock loads. Both member types may be designed for their respective dynamic loads by satisfying respective minimum requirements for structural member thickness and steel reinforcement content, and by satisfying stipulated equivalent static load conditions. Blast and shock load design usually proves more onerous than its peacetime design.

3.2 Design for Known Future Developments

Provision has been made along the route for developments. The loading requirements for such developments have been specified by the Urban Redevelopment Authority (URA), the Housing Development Board (HDB), and by the Port of Singapore Authority (PSA) and are incorporated into the Contracts.

While generally such loading is required to be supported directly by the NEL structures, provision is also made for separate piled foundations within approximately 5 m of the NEL structures for future developments.

The specified loading is applied on a building grid specified by the relevant Authority. To some extent, the Contractors may propose amendments to these grids to suit the station internal layout, to avoid transfer structures, and to optimise the structural design.

Except when explicitly agreed otherwise, any benefit of these developments to stability against flotation is ignored.

In the case of Dhoby Ghaut, the NEL Contractor is also constructing a commercial development, whose self-weight the Authority has accepted as contributing to stability against flotation. To provide for future redevelopment, with possible temporary removal of the weight of the development, a drainage blanket has been provided beneath the station to facilitate a temporary reduction in hydrostatic pressure underneath the base slabs, by pumping, and thereby to maintain stability against flotation during redevelopment.

At many stations allowance is made for future defined subway openings to be made to link future developments to the new stations. Generally, provision takes the form of trimming rebar around the openings plus modifications to the design to accommodate any redistribution of internal forces implied by making the future opening.

Allowance for future development is not limited to buildings, and provision has been made also for future transportation infrastructure.Similarly, provision is being made for future improvements to some of the main drainage canals: for example, the Bukit Timah Canal, for which piled foundations are installed by the NEL Contractor alongside the new NEL tunnels to support a future enlarged canal; the Whampoa Canal, with a new section being constructed

above the NEL bored tunnel; and the Eu-Tong Sen Canal, which is being upgraded in the Chinatown area.

3.3 Design for Other Future Developments

Provision for unknown future developments is made by requiring the structures to be checked for the effects of specified displacements. These correspond to the limits of deflections permitted under the Singapore "Code of Practice for Railway Protection" (CPRP) by future developments. As an alternative to designing for a fixed displacement, the designer may propose to cater for the worst credible loading resulting from future development(s). Because of the high stiffness of many of the structures, particularly the Civil Defence Stations, the worst credible loading is often less onerous than forcing a distortion of the structure to the 15 mm set out in the CPRP.

4.0 Trackwork

Because all of the NEL is in tunnel, a concrete track bed is appropriate. To ensure precise positioning of track fastenings and the accurate setting of the track to gauge, precast concrete sleepers set in in-situ concrete were selected. This arrangement has advantages in track-laying progress.

Grade 45 reinforced concrete sleepers, embedded in Grade 45 in-situ concrete at 700-mm centres, have been selected. The sleepers have exposed aggregate sides to promote bonding with the surrounding concrete. Using a reinforced rather than a prestressed sleeper allows the moulds to be stripped while the concrete is still green: there is no need to wait until the concrete becomes mature enough to be able to withstand the prestress loads. This makes it possible to simply produce an exposed aggregate finish along the sides of the sleepers.

The 60 kg/m rail section has been chosen for a combination of reasons — structural adequacy, electrical conductivity and availability. UIC Grade 900A with hardness not less than 260HB is used for all straight or curved track below 800-m radius. Rails on tighter radii are head-hardened to between 340 and 390HB. Rails are welded using the electric flash-butt welding process by an on-track machine, except within turnout limits, where thermit welding is accepted. Track gauge is 1435 mm.

Experience with the existing MRT has shown that the use of holding-down bolts to secure the base plates has two drawbacks. First, there is friction between the base-plate and the bolt, considerably reducing the designed resilience. Second, a bolted down arrangement does not lend itself to visual inspection of the bolt itself. The lower resilient pad puts high bending stresses into the bolts making them vulnerable to fatigue. Fatigue failure occurs just below the surface of the concrete and is usually not noticeable. Only when a number of adjacent failures have occurred will it be likely that visual inspection may discover the fault. But by then the track is in a dangerous condition and derailment is possible. Inspection must therefore be more scientific and ultra-sonic testing is called for. Even this arrangement has its problems, as the head of the holding-down bolt must be perfectly smooth and precisely perpendicular to the bolt for a reliable reading to be taken.

For the NEL, the important decision was made that base plates would not be connected to the sleepers with holding down bolts; instead, a clip with high fatigue endurance would be employed. A clip system has the immediate advantage that any clip that is broken or not fitting correctly can be seen visually or recorded by train-mounted video inspection.

A criterion was set for the level of re-radiated noise in properties in the vicinity of the MRT. 45dB(A) peak was selected as the most appropriate level, which took into account the noise environment in Singapore and the standards set by other countries in the world. A noise model was established and analyses carried out with various ground conditions and various levels of track resilience. It was determined that attenuation to meet the criterion could generally be achieved if two layers of resilience were provided within the support structure of the rail, thus avoiding the use of floating track systems where resilience is provided beneath a supported concrete slab. A pad beneath the rail and second pad beneath the between the base plate and

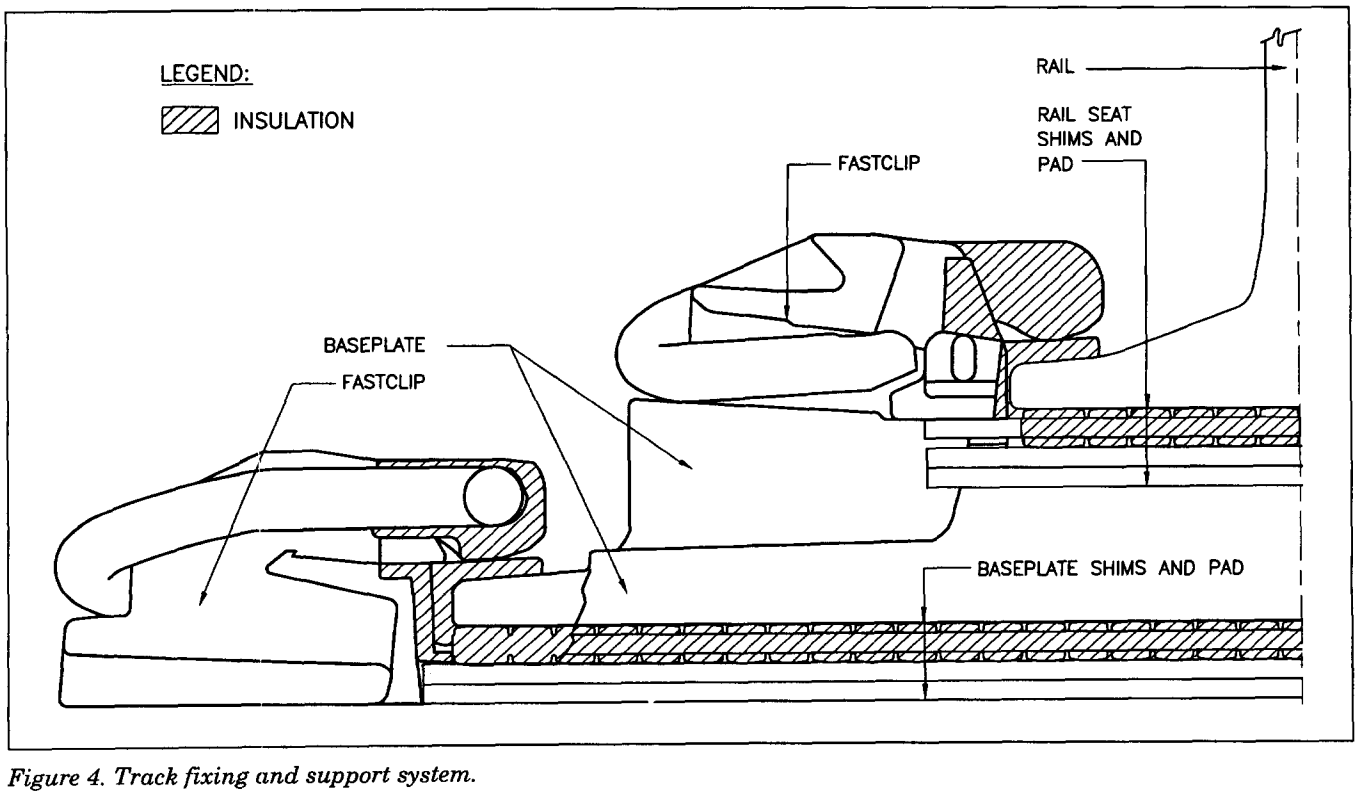


Figure 4. Track fixing and support system.

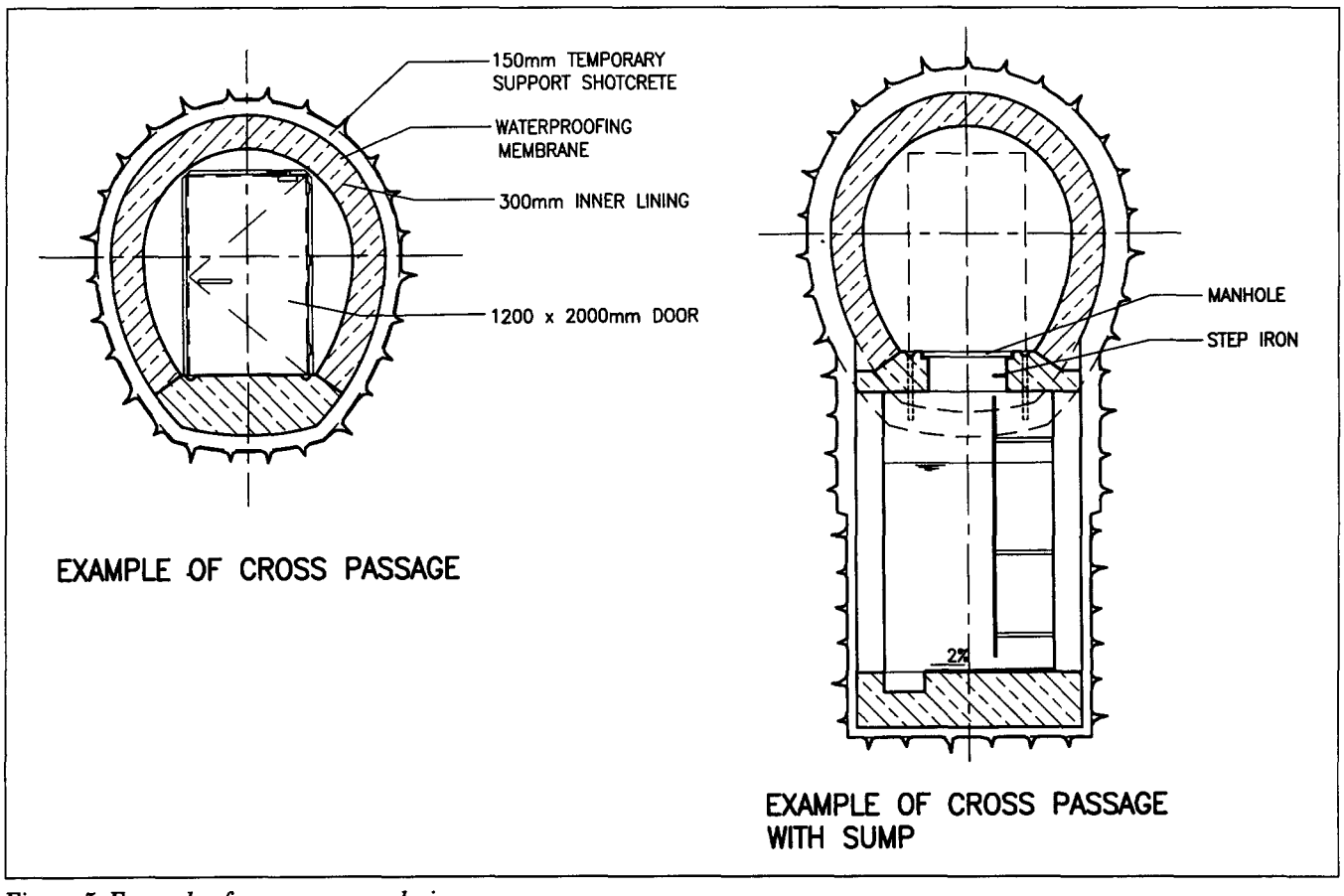


Figure 5. Example of cross-passage design.

its support were specified to produce an overall resilience of 37kN/mm per metre of rail length and a maximum rail deflection of 3 mm.

The selected track system consists of a cast base plate supported on a resilient pad and fixed to the sleeper by FastClips by Pandrol with the Pandrol shoulder cast into the sleeper. The rail is attached to the base plate by a second FastClip. The resultant arrangement is termed Double FastClip. Current control is provided by insulation pieces fitted around both layers of FastClips, and by both resilient pads. The arrangement is shown in Figure 4.

Both rail pads are manufactured from natural rubber and the designed resilience is shared approximately equally between the two pads. Although the two pads are made from the same material, the stiffness of the lower pad is engineered to match the stiffness of the upper pad by the adoption of a castellated surface. High-density polypropylene shims are inserted beneath each pad to provide some tolerance in level and to permit future adjustments as necessary. The track will be constructed with 20 mm of shims in place. The 20 mm will consist of two 5-mm shims under the rail, and two 5-mm shims under the baseplate, as shown in Figure 5. Any future adjustment to the rail level will be by replacement of the 5-mm shims with thinner ones. A minimum of 3 mm of shims is required under both the rail and the baseplate, so there is the potential for changing the rail level by up to 14 mm. The FastClip is designed to work in this range.

In addition to the electrical insulation of the track fittings, the rail is supported so that the clearance beneath it to the track concrete is 90 mm. This clearance was chosen to minimise the likelihood of detritus accumulating and forming a route for stray currents to find a path to earth. The insulation level (track to ground resistance) specified for new track is 10 ohms for 1 km of track under damp conditions.

Below the sleeper level, some 250 mm below the surface of the trackbed concrete, there is a welded reinforcement mesh having 10-mm bars at 200-mm centres in both directions. This mesh is there purely for stray current purposes (providing 1100 mm^2 of conductor) and is welded at laps into continuous lengths of 100 m. Each 100-m length is electrically separate from the next.

5.0 Tunnel Design

The North East line has a total route length of about 20 km, of which some 11 km are being constructed in bored tunnel. Almost all of the bored tunnel is lined using precast concrete segments. The minor exceptions to this are 50 linear metres of tunnel constructed using a temporary shotcrete lining and 200 linear metres constructed using spheroidal graphite iron (SGI) segments.

The design of the precast concrete tunnel linings for the North East Line is described in some detail in the paper by Doran et al. (1999). Table 3 provides a summary of tunnel linings used.

Comparing the linings used for the earlier phases of the Singapore MRT system with those used for the North East line, a major change has been in the length of the rings. The North East line rings have varied from 1.2 m to 1.5 m in length, compared with the 1.0 m used for the earlier phases. This has had a significant impact on the design of the crosspassages between the running tunnels. The requirement for crosspassages is specified to follow NFPA 130.

The Authority has encouraged designers to make the crosspassages as small as practicable; the minimum dimensions are essentially dictated by the necessary size of the crosspassage door. This door has a minimum width of 1.2m. With the longer tunnel lining segments, it has been possible to design the opening so that only a very limited number of segments are removed to make the opening, so greatly simplifying the design of the lintel and jambs framing the

door. An example of the resulting crosspassage design is shown in Figure 5.

6.0 Temporary Works Design

6.1 Selection and Design

As discussed above, almost all of the tunnels have or are being driven using Earth Pressure Balance Shields and supported by a one-pass reinforced concrete lining. Temporary works design issues for the bored tunnels have therefore been limited to items such as cross-passages and shafts.

In contrast, a wide variety of temporary works schemes were proposed for the construction of stations and cut and cover tunnels. These are summarised in Tables 1 and 2. In the northern section of the works, between Hougang and Punggol stations, most of the construction was carried out in open cut. This area was largely undeveloped before NEL construction, and the excavation was carried out mainly in Old Alluvium.

The area between Boon Keng and Kovan is also predominantly in the Old Alluvium, except for the buried ridge of Bukit Timah granite at Serangoon. This area was developed, predominantly by mass housing blocks constructed since the 1960s by the Housing Development Board (HDB). These blocks are typically 6 to 12 stories high and constructed of reinforced concrete on piled foundations. The temporary works systems used in this area varied from Diaphragm Walls (at Boon Keng, Potong Pasir, and Kovan) to soldier piles and lagging (at Serangoon and Woodleigh).

The section of the line between China Town and Farrer Park Stations includes many of the areas where deep deposits of the soft and highly compressible Singapore Marine clay were encountered. This section also includes many of the older areas of Singapore, such as China Town and Little India, which are conservation areas. The characteristic structure in these older areas consists of 2- to 4-storey shophouses built between 1880 and 1950. The shophouses are typically built of brick or have a reinforced concrete frame with brick infill. The original foundations are generally strip or pad footings, although in areas of soft ground the ground was often reinforced by a mass of short timber (bakau) piles. As part of the conservation of these areas, many of the shophouses have been renovated; in some cases the renovation has involved underpinning using mini-piles. The four stations in the deep marine clay (Farrer Park, Little India, Clarke Quay and Chinatown) have all been constructed with the main excavations supported by strutted diaphragm or secant pile walls. In the areas where the marine clay is the deepest, jet grouted rafts have been provided at the base of the excavation to limit wall deflection.

The three remaining stations (Dhoby Ghaut, Outram Park and World Trade Centre) have mainly involved excavation through the weathered sedimentary rocks of the Jurong Formation. Locally, up to 10 m of soft organic clay has been encountered above the Jurong Formation. Generally, the system of support used has been soldier piles with shotcrete lagging; this has been supplemented with sheetpiles through soft clay where present. However, two of the stations—Dhoby Ghaut and Outram Park—involve construction close to existing Mass Rapid Transit tunnels and stations. For the sections of the work adjacent to the operating system, secant piles or diaphragm walls were specified; in both cases, the contractors have selected and installed secant pile walls.

Following are the main steps in the design of the temporary walls for the station and cut-and-cover tunnels:

1. Select toe depth required to provide stability (as required in BS8002).
2. Carry out seepage analysis.
3. Carry out wall analysis to assess moments, shears and deflections of the wall.
4. Assess strut loads.
5. Predict ground movements due to wall installation, wall deflection and consolidation.
6. Assess impact on buildings and utilities due to ground movements.

This raised a number of issues, which are discussed below.

6.2 Temporary Works Design Issues

Design of the temporary retaining walls and struts or anchors is required to be carried out to BS 8002. However, in the design a major issue has been the appropriate design

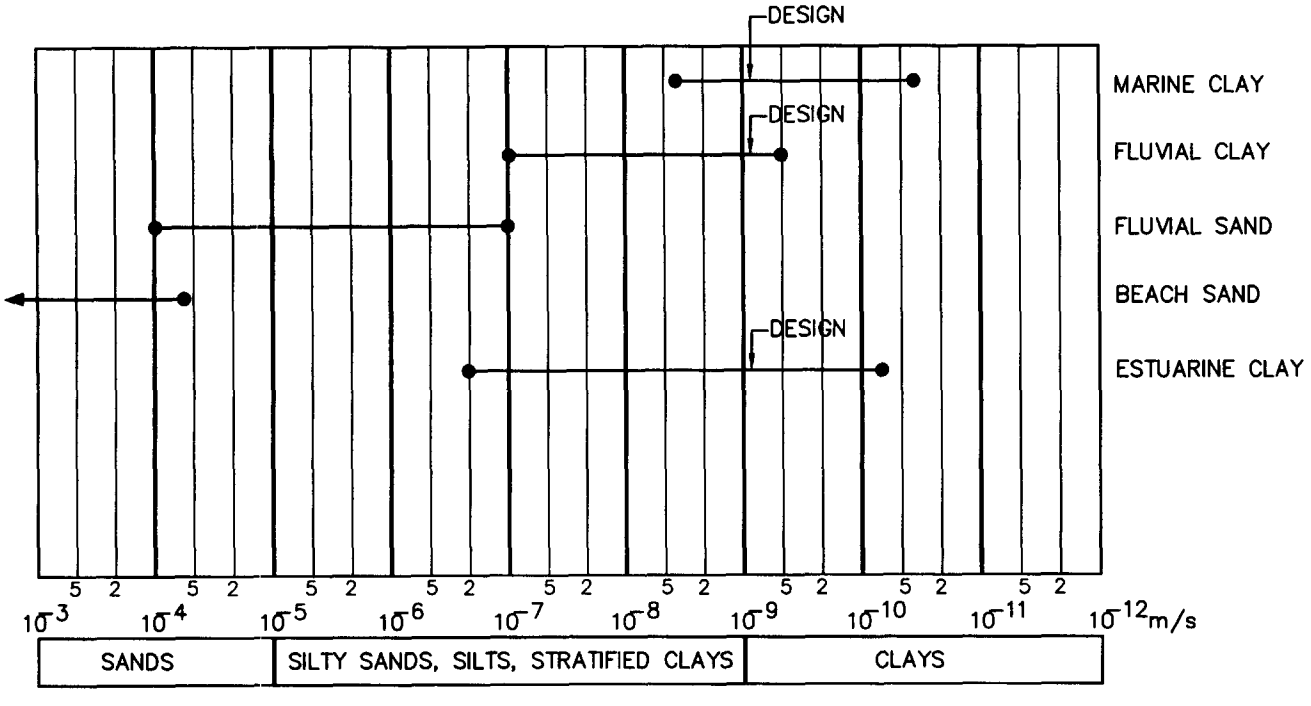


Figure 6. Typical ranges of permeability for Kallang Formation soils.

Table 3. A summary of the bored tunnel linings used on the North East Line.

Contract	C703	C704	C705	C706	C708	C710
Contractor	Samsung	Hyundai	Kumagai-sembawang -Mitsui JV	Hyundai-Zublin JV	Nishimatsu-Lum Chang-Bachy Soletanche JV	Shimizu-Dilingham-Koh Brothers JV
Designer	PBMM	Geoconsult	Babtie	Geoconsult	Maunsells	By contractor
Approximate Total Tunnel Length	2.7 km	5.3 Km	3.7 Km	3.3 Km	3 km	5 km
Lining Manufacturer	Hume Industries (M) Berhad	Redlands Precast (HK)	Sembawang Colebrand JV	ACPI	ACPI	Hume Industries (M) Berhad
Country Of Manufacture	Malaysia	Singapore	Singapore	Malaysia	Malaysia	Malaysia
Lining Thickness	250 mm	250mm	250 mm	250 mm	275 mm	250 mm
Segment Length	1.2 m	1.5 m	1.2 m	1.5 m	1.2 m	1.2 m
Segmentation	5 no. 67.5° segment + 1 bolt Key	3 no. 72° segment + 2 no. 65.6° + 1 Key	5 no. 60° + 1 no. 45° segment + 1 bolt Key	5 no. 72° segment + 1 Key	5 no. 68° segment + 1 bolt Key	3 no. 72 ° + 2 no. 64.5 ° segment + 1 bolt Key
Reinforcement Quantities	0.60% - 0.83%	0.48% – 1.44%	0.54%	0.64% -1.07%	0.41 %	0.54 % - 0.8 %
EPDM Gaskets	Phoenix M38541a	Daetwyler Type 86-259	Heinke CS003	Phoenix M38544	Phoenix M38541a	Heinke CS004
Hydrophilic Gaskets	AOI Bond Ring W N-4 (4 x expansion)	Kasei DST-0518-35-I (4 x expansion)	Kasei DSS-0220 (8 x expansion)	AOI Bond Ring W N-4 (4 x expansion)	Kasei DST-0520-35-I (4 x expansion)	Kasei DST-04518-30-I (4 x expansion)
Radial Bolts	Curved Bolts 2 no. M22	Straight Bolts 2 no. M24	Curved Bolts 2 no. M24	Straight Bolts 2 no. M 22	Curved Bolts 2 no. M24	Curved Bolts 2 no. M24
Circumferential Bolts	Curved Bolts 16 no. M22	Straight Bolts 10 no. M24	Curved Bolts 24 no. M24	Straight Bolts 10 no. M22	Curved Bolts 21 no. M24	Curved Bolts 16 no. M24
Circumferential Joint Packings	Kaubit Pads	Bituminous felt	None	Kaubit Pads	None	Kaubit pads
Radial Joint Type	Block Joint	Block Joint	Double Convex	Block Joint	Block Joint	Block Joint

parameters to use, and therefore the soil and groundwater pressures to use for design.

The weakest soils in Singapore are the marine and fluvial soils of the Kallang Formation. However, these are the soils for which it is simplest to establish suitable design parameters for temporary works. As shown in Figure 6, the Kallang Formation divides neatly into the clays of very low permeability, and sands of much higher permeability. For the clays, total stress parameters may be used. The undrained cohesion typically increases linearly with depth as the clays are near normally consolidated. For the marine clay, the undrained strength for design is often taken as 0.25 x vertical effective stress.

Establishing suitable design parameters for the weathered rocks of the Jurong Formation, Bukit Timah Granite and Old Alluvium is less simple than for the Kallang Formation soils. It has been common in Singapore to use undrained parameters for the weathered rocks. It has also been common to derive those parameters, even for complex computer analyses, by extrapolation from the results of Standard Penetration Testing. However, as shown in Figure 7, the permeability of these materials typically varies from that of a silty sand to that of a clay. The use of undrained parameters in materials with this range of permeability, and in often highly fractured rocks, is clearly questionable. For the North East Line, the design of the temporary works has typically been carried out either with effective stress parameters, or checking for the worst case between effective and total stress parameters.

For the Old Alluvium, which is typically intact and unfractured, effective stress parameters were based on the results of consolidated triaxial testing. To capture the effects of weathering near the surface of the deposit, the Old Alluvium was typically divided into bands based on the results of Standard Penetration Testing.

Strength parameters for the weathered rocks of the Jurong Formation could not be derived directly from Triaxial testing. The highly and moderately weathered rocks are typically highly fractured, due to the effects of both folding and weathering. The beds dip steeply, typically at 50° to 80°, and adjacent beds often vary greatly in degree of both fracturing and weathering. Strength parameters for design were typically derived using the methods outlined in Hoek and Brown 1997.

In the weathered rocks, the design of struts and anchors were typically checked using trapezoidal "apparent earth pressure" diagrams. The application of these diagrams (Peck, Hanson and Thorburn 1974) under Singapore conditions was reviewed by Hwang et al (1987).

The assessment of the effects on adjacent buildings is required to be carried out using the methods outlined in Mair, Burland and Taylor. In applying these methods and criteria to buildings in Singapore, there have been a number of issues related to the current state of the buildings, particularly the old shophouses common in the central areas of Singapore.

Many of the older shophouses were built in areas underlain by deep deposits of marine clay. The low strength and high compressibility of this clay means that many of the shophouses have suffered significant settlement over the years. As a result, a number of shophouses were found that showed existing damage exceeding the "slight" damage criterion. In these cases the criterion could not be applied, and the contractor was required to ensure the safety of the occupants and passers-by during his work. This was typically done by shoring the buildings.

Extensive work to restore (in conservation areas) or redevelop (in other areas) old shophouses has been carried out in recent years. In restoration or redevelopment, it is common to pile the structures through the marine clay and into hard strata. However, the design of the new foundation system has frequently resulted in buildings that are unusually sensitive to settlement. This can occur in two ways:

- The centre of the structure is extended upwards, but the façade is maintained for reasons of conservation. The new central columns are piled, but the façade (including the traditional Singapore 5-foot way) is left on the original shallow foundations
- One or more of a terrace of shophouses is demolished and a new, piled, structure constructed. However, the new building is cast up against the rough brickwork of

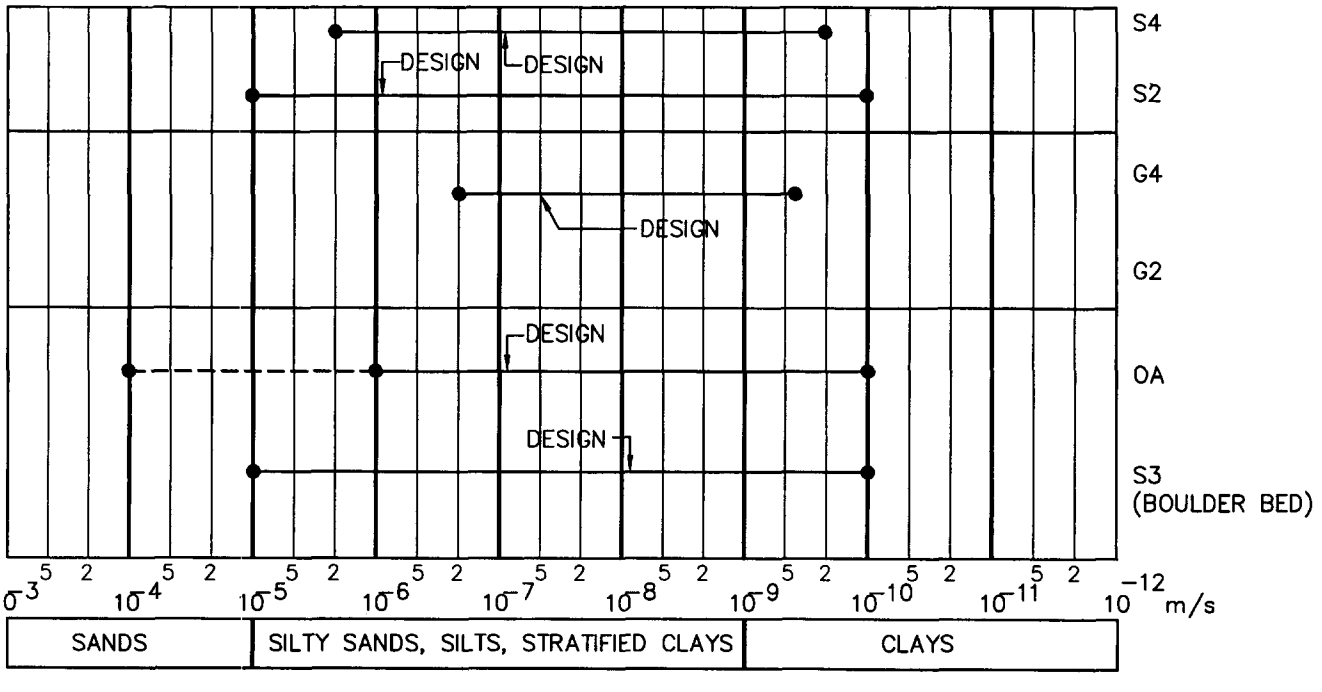


Figure 7. Typical ranges of permeability for weathered rocks in Singapore.

the party wall of the adjoining shophouses, which are left on their original shallow foundations.

The sensitivity of such structures to settlement is exceptional. In one case of a building near the NEL, a 15-mm crack in the structure was caused by just 11 mm of settlement. In this case the building, a church, had to be evacuated and underpinned to make it safe for the NEL excavation to continue.

Concluding Comments

In a paper describing various facets of the civil design of a major, underground, design/build transit project, it is difficult to draw specific conclusions. Indeed, in the case of the North East line, which is still under construction, it would be tempting fate to come to specific conclusions. However, it can be said that the overall project is progressing satisfactorily. The design of the project has been significantly complicated by the many provisions which have had to be made for the future construction of other rail and road projects, future developments and drainage and sewerage schemes. However, this can be seen as another example of the high level of forward planning necessary in land scarce Singapore.

References

Boscardin, M. D. and Cording, E. G. 1989. Building response to excavation induced settlement. *J. Geot. Eng.*, ASCE, **115**.

Burland, J. B.; Broms, B.; and De Mello, V. F. B. 1977. Behaviour of foundations and structures. SOA Report. Session 2. *Proc. 9th Int. Conf. S.M.F.E, Tokyo, Vol 2*, 495-546.

Curtis, D. J. 1976. Discussion on the circular tunnel in elastic ground. *Geotechnique* **26** (1).

Doran, S. R.; Copsey, J. P.; and Poh, S. K. 1999. Developments in Precast Tunnel Lining Design for the Singapore M.R.T. *Proc. International Conference on Rail Transit, Assoc. Consult. Eng. Singapore, Vol 1*, 337–348.

Gupta, A.; Rahman, A.; Wong, P. P.; and Pitts, J. 1987. The Old Alluvium of Singapore and the extinct drainage system to the South China Sea. *Earth surface process and landforms* 12, 259–275.

Hoek, E. and Brown, E. T. Practical estimates of rock mass strength. *Int. J. of Rock Mech.*

Hwang, R., Quah, H.P. and Buttling, S. Measurements of strut forces in braced excavations. *Proc. Singapore Mass Rapid Conference*, 239–244.

Mair, R. J., Taylor R. N. and Burland, J. B. 1996. Prediction of ground movements and assessment of risk of building damage due to bored tunnelling. *Proc. Int. Symposium "Geotechnical aspects of underground construction in soft ground"*, 713-718.

Muir Wood, A. M. 1975. The circular tunnel in elastic ground. *Geotechnique* **25** (1).

Peck, R. B., Hanson, W. and Thorburn, T. 1974. *Foundation Engineering. 2nd Ed.* New York: Wiley.

Singapore Public Works Department. 1976. *Geology of the Republic of Singapore.* Singapore: Public Works Department.